

- 6 An a.c. power supply is connected to a resistor R , as shown in Fig. 6.1.

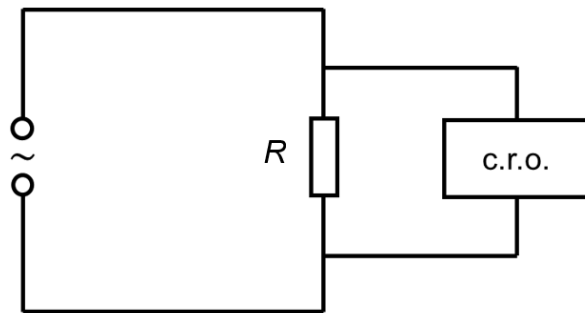


Fig. 6.1

A cathode ray oscilloscope (c.r.o.) is used to show the potential difference (p.d.) across R . The screen of the c.r.o. displays the variation with time of the p.d. across R , as shown in Fig. 6.2.

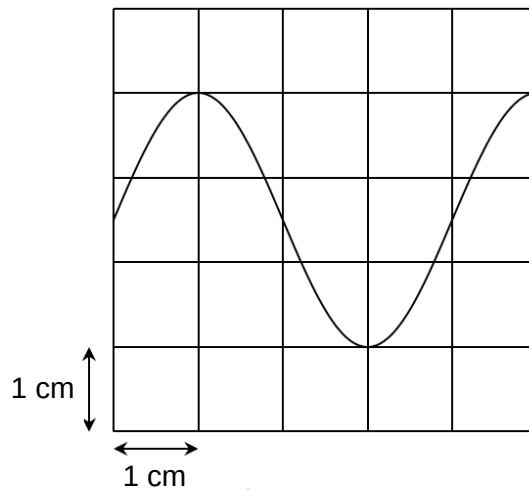


Fig. 6.2

- (a) Explain, by reference to direct current, what is meant by the *root-mean-square* (r.m.s.) value of an alternating current.

The r.m.s. value of an alternating current is the value of the steady current that will dissipate heat at the same average rate as the alternating current in a resistive load [B1]

..... [2]

- (b) The voltage V of the power supply is given by the expression

$$V = 6.0 \sin(100\pi t)$$

V is measured in volts and the time t is measured in seconds.

Determine the Y-gain and time-base of the c.r.o.

12.0 V corresponds to 3 cm.

$$12 \div 3 = 4.0 \text{ V/cm} \quad [\text{A1}]$$

$$\frac{2\pi}{T} = 100\pi$$

$$T = 20 \text{ ms}$$

20 ms corresponds to 4 cm.

$$20 \div 4 = 5 \text{ ms} \quad [\text{A1}]$$

Y-gain = V / cm

time-base = ms / cm
[2]

(c) A diode is then connected in series with the resistor R , as shown in Fig. 6.3.

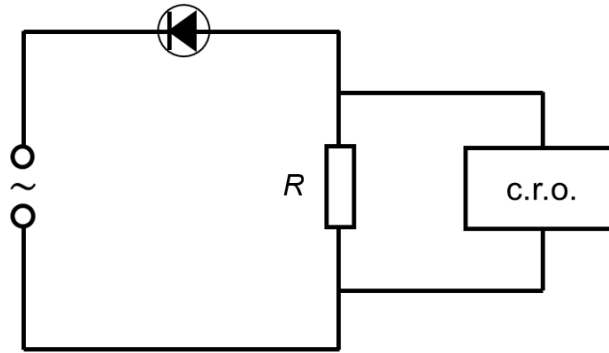


Fig. 6.3

- (i) Sketch in Fig. 6.4, for 2 periods of the alternating p.d., the variation with time of the power P dissipated in R when R is $20\ \Omega$, where T is the period. Indicate the peak power value in your sketch. [2]

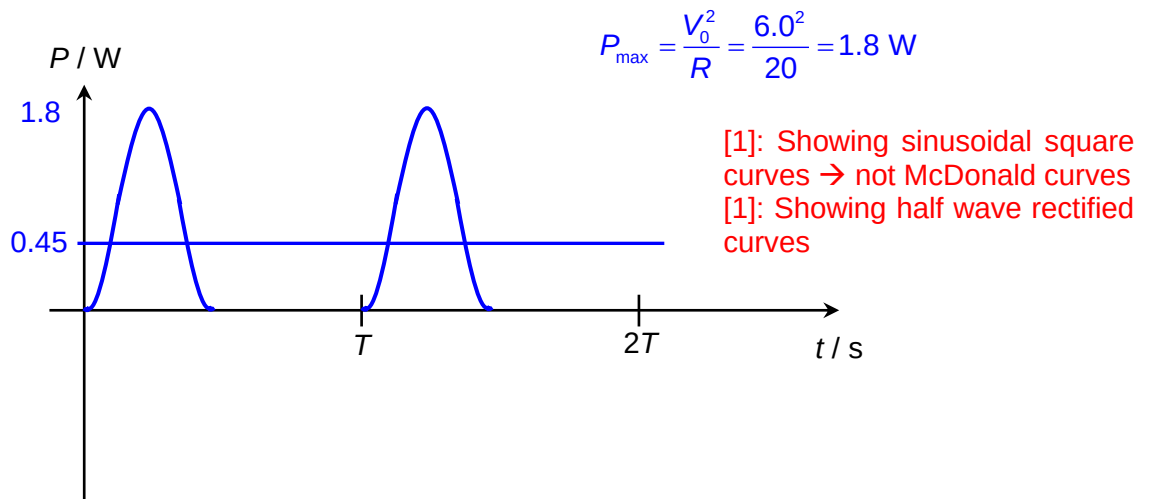


Fig. 6.4

- (ii) Draw a line in Fig. 6.4 to represent the average power dissipated in R . Indicate its value on the y-axis. [1]
- (iii) Calculate the root-mean-square current in R .

$$\begin{aligned}\langle P \rangle &= I_{\text{rms}}^2 R \\ 0.45 &= I_{\text{rms}}^2 (20) \\ I_{\text{rms}} &= 0.15\text{ A}\end{aligned}$$

root-mean-square current = A [2]

[Total: 9]

7 Fig. 7.1 shows what is observed when a parallel beam of electrons, accelerated by a potential